

✓ Lab 1 — Exercise: Notebooks & Markdown

ITCS 3162 — Introduction to Data Mining

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Complete the exercises below. For each task:

- **Code tasks** → fill in the code cell
- **Written answers** → edit the markdown cell directly (look for **YOUR ANSWER:** prompts)

When you're done, run **Kernel → Restart & Run All** to make sure the whole notebook executes top-to-bottom without errors, then download the `.ipynb` and submit via Canvas.

Disclaimer: I used AI to assist with code generation and understanding.

✓ Exercise 1 — Running cells (5 pts)

Run the cell below. Then **modify** it so that it prints your full name and your major.

```
# TODO: change the strings to your name and major
name = "Ruhan Rahman"
major = "Data Science & Business Analytics"
print(f"Hi, I'm {name} and I'm studying {major}.")
```

Hi, I'm Ruhan Rahman and I'm studying Data Science & Business Analytics

✓ Exercise 2 — Variable persistence (10 pts)

Write code in the next cell that creates two variables, `a` and `b`, with any numeric values you like. Then in the cell after that, print their sum, difference, and product on three separate lines.

```
# TODO: define a and b
a = 6
b = 9
```

```
# TODO: print a+b, a-b, a*b on separate lines
print(a+b)
print(a-b)
print(a*b)
```

```
15
-3
54
```

Exercise 3 — Markdown formatting (15 pts)

Double-click this cell and replace the placeholder content below with:

1. A level-2 heading with the text "About me"
2. A bulleted list with three items: your favorite class, a hobby, and a goal for this course
3. One sentence containing at least one **bold** and one *italic* word
4. An inline code reference to `pandas`
5. A link to a website you find useful

About me

- **Favorite class:** Data Science Studios
- **Hobby:** Urban hiking
- **Goal for this course:** To learn a lot :)

I *hope* that I learn a **lot** in this class this summer.

I've been practicing using `pandas` in Python.

Here is a website that I find useful: [DataCamp](#).

Exercise 4 — Math notation (10 pts)

YOUR ANSWER: $y=mx+b$

In this equation, m represents the slope of the line.

Exercise 5 — A small table (10 pts)

In the cell below, create a markdown table listing **three data mining tasks** we'll cover this semester. Columns: **Task**, **Type** (supervised / unsupervised / association), and **Example use case**.

YOUR ANSWER:

Task	Type	Example use case
Classification	Supervised	Predicting whether a credit card transaction is fraudulent or not.
Clustering	Unsupervised	Segmenting customers into groups based on purchasing behavior.
Association Rule Mining	Association	Market basket analysis to find that customers who buy diapers also fre

Exercise 6 — Reflection (10 pts)

In 2–4 sentences, describe one thing you found confusing about notebooks and one thing you find useful compared to writing a regular Python script.

YOUR ANSWER: One thing that can be confusing about notebooks is remembering to switch between markdown and code cells when appropriate. One advantage compared to regular Python scripts is the ability to write and run code in blocks, which really helps with organization.

Submission checklist

- ☐ Replaced your name and date at the top
- ☐ All code cells run without errors
- ☐ All **YOUR ANSWER** prompts have been replaced
- ☐ Ran **Restart & Run All** before exporting
- ☐ Downloaded as **.ipynb** and uploaded to Canvas

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